



IILM University

 **AICTE EduSkills Virtual Internship Capstone**

AI-ML Virtual Internship

Heart Failure Risk Prediction Web Application

An End-to-End Machine Learning Solution for Cardiovascular Risk Stratification & Clinical Decision Support

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Program : B.Tech-CSE

Semester: 5th

IILM University, Greater Noida

Organization: Eduskills Academy

Internship Background & Objectives



Collaborative Initiative

Organized by AICTE and EduSkills Foundation, supported by Google for Developers to build industry-ready talent in AI and Machine Learning.



8-Week Structure

Comprising 4 weeks of structured learning on Google Developer platforms followed by 4 weeks of capstone project engineering and mentorship.

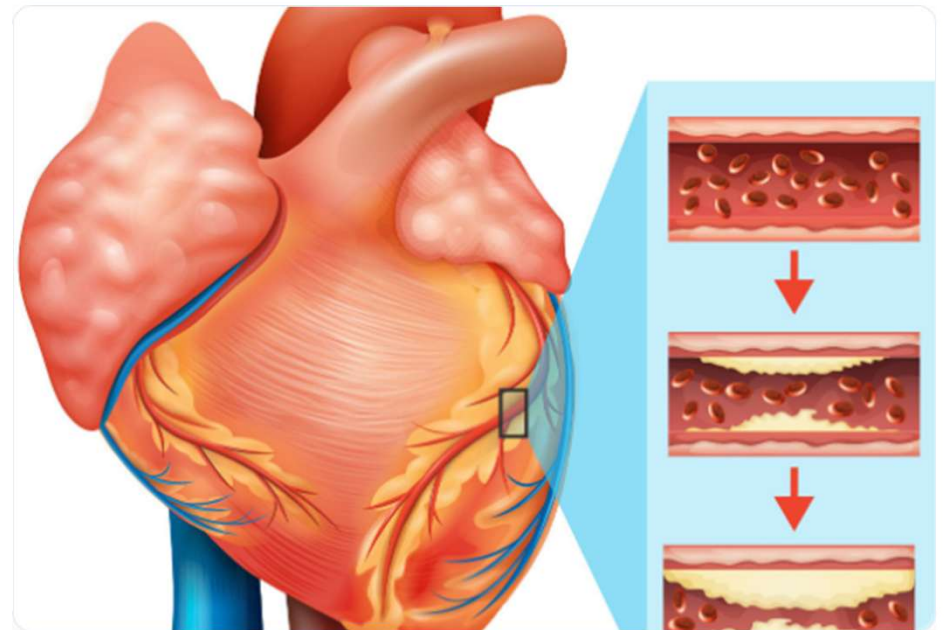


Core Outcome

Successfully achieved Grade-O (Outstanding) by delivering an end-to-end Machine Learning model deployed via a responsive Flask web application.

Problem Statement & Clinical Context

- ❗ **Global Health Crisis:** Cardiovascular Diseases (CVDs) cause approximately 17.9 million deaths annually, accounting for 31% of global mortality.
- 🔗 **Diagnostic Bottlenecks:** Conventional diagnostic methods rely on manual interpretation of multiple physiological markers, leading to delayed early-stage interventions.
- ⚙️ **Machine Learning Opportunity:** Automated analysis of multi-attribute patient records can quickly highlight high-risk individuals and improve patient outcomes.



Kaggle Heart Failure Dataset Features

Feature Name	Type	Description & Standard Units
Age, Sex	Numeric / Categorical	Patient age in years; Biological sex (M/F)
ChestPainType	Categorical	Atypical Angina (ATA), Non-Anginal (NAP), Asymptomatic (ASY), Typical Angina (TA)
RestingBP, Cholesterol	Numeric	Resting Blood Pressure (mm Hg); Serum Cholesterol (mg/dl)
FastingBS, RestingECG	Binary / Categorical	Fasting Blood Sugar > 120 mg/dl (1/0); Normal, ST-T abnormality, LVH
MaxHR, ExerciseAngina	Numeric / Binary	Maximum Heart Rate achieved (60–220 bpm); Exercise-induced angina (Y/N)
Oldpeak, ST_Slope	Numeric / Categorical	ST depression during exercise; Peak exercise ST segment slope (Up/Flat/Down)

Data Preprocessing & Serialization



1. Dummy Encoding

Categorical attributes (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope) transformed using One-Hot Encoding (`pd.get_dummies`).



2. Standard Scaling

Numeric features scaled using `StandardScaler` wrapped in `ColumnTransformer` (`remainder='passthrough'`) to prevent data leakage.



3. Pipeline Persistence

Serialized key assets (`.joblib`): trained model, transformer scaler, original feature order, and encoded column names for web runtime reindexing.

Model Training & Evaluation Metrics

85.87%

Accuracy Score

86.27%

Precision

88.00%

Recall Sensitivity

87.13%

F1-Score



Optimized Classifier: Logistic Regression trained with L2 regularization (`max\_iter=1000`) on stratified train/test split (80:20).



High Recall Priority: Achieved an 88.0% recall rate, minimizing false negatives to ensure high-risk cardiac patients are accurately flagged.

Confusion Matrix (Test Set)

TN: 70

FP: 14

FN: 12

TP: 88

Flask Application & Web Backend



Flask Framework (`app.py`): Implements lightweight HTTP endpoints (`/` and `/predict`) to serve model predictions in real time.



Dynamic Preprocessing: Receives form payload, builds single-row DataFrame, applies One-Hot Encoding, reindexes columns against `encoded\_features`, and scales via `joblib` scaler.



Probability Risk Output: Extracts class 1 failure probability (`predict\_proba`) to compute real-time risk percentage.

Data Analytics Dashboard Concept

Interactive User Interface & Gauge



SVG Gauge Needle

Dynamic semi-circle gauge animated via CSS transforms, mapping probability (0-100%) to needle rotation (-90° to $+90^\circ$).



Risk Categorization

Automatically segments results into **Safe** (<30%), **Serious** (30–70%), and **Critical** ($\geq 70\%$) risk tiers with color-coded alerts.



Clinical Advice

Generates tailored, actionable lifestyle and medical recommendations based on predicted risk severity inside the response modal.

End-to-End User Experience Workflow



Clinical Value & Key Takeaways

-  **Early Detection Tool:** Provides immediate risk assessment before intensive clinical testing.
-  **Robust Validation:** Client-side HTML5/JS input validation prevents invalid medical parameter submissions.
-  **Scalable Deployment:** Lightweight Flask backend suitable for cloud hosting on platforms like AWS, Render, or Heroku.



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Thank You!

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